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Door latch device and actuator

Abstract

A door latch device includes: a rotating body; first and second levers; a lock mechanism on which an unlocking operation is performed and an open link is disposed in an unlocked position; and a latch mechanism. Further, the lock mechanism includes a lock lever which causes the open link to rotate when being operated by rotation of the rotating body, the rotating body is provided with two tilted walls so as to move the open link to the unlocked position when being rotated in one side from a reference position while the open link is disposed in a locked position, keep the open link in the unlocked position when being rotated in another side to the reference position while the open link is disposed in the unlocked position, and move the open link to the locked position when being further rotated in the another side.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of International Application No. PCT/JP2020/002710, filed on Jan. 27, 2020 which claims the benefit

of priority of the prior Japanese Patent Application No. 2019-204785, filed on Nov. 12, 2019, the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to a door latch device and an actuator.

(2) A latch mechanism and a lock mechanism are normally disposed on a door latch device mounted on a vehicle. The latch mechanism is configured to maintain a closed state of a door with respect to a vehicle main body by engaging with a striker disposed on the vehicle main body, for example. The lock mechanism is configured to be switched between an unlocked state for releasing an engaging state of the latch mechanism in a case in which an operation of opening the door is performed with a door handle, and a locked state for maintaining the engaging state of the latch mechanism by invalidating the operation for the door handle. As such a type of door latch device, there has been developed a door latch device configured to perform an operation of releasing the engaging state of the latch mechanism and an operation of switching the locked state to the unlocked state by driving of a single actuator. More specifically, a rotating body, is rotated by a motor, the rotating body including a projecting part for causing the latch mechanism to operate disposed on one end face and a projecting part for causing the lock mechanism to operate disposed on an outer peripheral surface, and the projecting parts are respectively caused to work on the latch mechanism and the lock mechanism to successively perform an engagement release operation for the latch mechanism and an unlocking operation for the lock mechanism (for example, refer to Japanese Patent No. 6213927).

SUMMARY

(3) There is a need for providing a door latch device and an actuator by which an increase of the size thereof can be prevented.

(4) Accord in to an embodiment, a door latch device includes: a rotating body to be rotated and driven with respect to a case; a first lever and a second lever disposed to be rotatable with respect to the case, being configured to be operated by a cam part disposed on the rotating body; a lock mechanism on which an unlocking operation is performed so that an open link is arranged at an unlocked position in a case in which the first lever rotates; and a latch mechanism on which a release operation is performed in a case in which the second lever rotates. Further, the cam part is disposed only on one end face of the rotating body, and a first cam to operate the first lever and a second cam to operate the second lever are individually disposed in the cam part, the lock mechanism includes a lock lever to be operated in accordance with rotation of the rotating body to rotate the open link, and the rotating body includes two tilted walls to cause the open link to move to the unlocked position via the lock lever in a case in which the rotating body rotates in one direction from a reference position in a state in which the open link is arranged at a locked position, maintain the open link at the unlocked position in a case in which the rotating body rotates in another direction to return to the reference position from a state in which the open link is arranged at the unlocked position, and cause the open link to move to the locked position in a case in which the rotating body further rotates in the another direction from the reference position.

(5) According to an embodiment, an actuator includes: a rotating body configured to be rotated and driven with respect to a case; and a first lever and a second lever that are disposed to be able to rotate with respect to the case, and configured to be operated by a cam part disposed on the rotating body. Further, the cam part is disposed only on one end face of the rotating body, and a first cam configured to operate the first lever and a second cam configured to operate the second lever are individually disposed in the cam part, and a starting end of a cam surface of the second cam is disposed to be closer to an outer peripheral side than a starting end of a cam surface of the first cam is, and an increasing ratio of an outer diameter of the cam surface of the second cam is set to be smaller than an increasing ratio of an outer diameter of the cam surface of the first cam.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a door latch device according to an embodiment of the present disclosure viewed from an outer side and an obliquely rear side of a vehicle;
- (2) FIG. 2 is a perspective view of the door latch device viewed from the outer side and an obliquely front side of the vehicle;
- (3) FIG. 3 is a side view illustrating an inner part of the door latch device;
- (4) FIG. 4 is a perspective view of a latch mechanism;
- (5) FIG. 5 is a perspective view of a lock mechanism viewed from an obliquely inner and rear side;
- (6) FIG. 6 is a perspective view of the lock mechanism viewed from an obliquely outer and front side;
- (7) FIGS. 7A and 7B illustrate a cam wheel,
- (8) FIG. 7A is an end face diagram;
- (9) FIG. 7B is a perspective view;
- (10) FIGS. 8A and 8B are a diagram illustrating an operation of the lock mechanism viewed from an inner side of the vehicle when the cam wheel normally rotates;
- (11) FIG. 8A is a diagram illustrating a basic state in which the cam wheel is at a reference position;
- (12) FIG. 8B is a diagram illustrating a state in which the cam wheel normally rotates from the reference position by substantially 100 degrees;
- (13) FIGS. 9A, 9B, 9C, and 9D are diagrams illustrating the operation of the lock mechanism viewed from the outer side of the vehicle when the cam wheel reversely rotates and normally rotates;
- (14) FIG. 9A is a diagram illustrating a basic state in which the cam wheel is at the reference position;
- (15) FIG. 9B is a diagram illustrating a state in which the cam wheel reversely rotates from the reference position by substantially 40 degrees;
- (16) FIG. 9C is a diagram illustrating a state in which the cam wheel normally rotates by substantially 40 degrees from the state of FIG. 9B; and
- (17) FIG. 9D is a diagram illustrating a state in which the cam wheel normally rotates by substantially 40 degrees from the state of FIG. 9C.

DETAILED DESCRIPTION

- (18) With the door latch device in the related art, the release operation for the latch mechanism and the unlocking operation for the lock mechanism can be performed by a single motor, so that there are advantages such that the number of components can be reduced, for example. However, components such as a lever to be engaged with the projecting part to be operated are required to be disposed in a region opposed to the end face of the rotating body and a region opposed to the outer peripheral surface of the rotating body, so that the size of the device may be increased.
- (19) The following describes a preferred embodiment of a door latch device according to the present disclosure in detail with reference to the attached drawings.
- (20) FIG. 1 to FIG. 3 are diagrams illustrating the door latch device according to the embodiment of the present disclosure. Although not illustrated in the drawings, the door latch device exemplified herein is mounted on a side door disposed on the right side of a four-wheeled automobile, and performs opening/closing control for the side door by changing an engaging state with a striker disposed on a vehicle main body in accordance with an opening/closing operation with a door handle or a locking/unlocking operation with a key.
- (21) As illustrated in FIG. 1 and FIG. 2, in a door latch device **10**, a latch member **12** for latching the striker is disposed in the rear of a striker entry groove **14**. The latch member **12** is part of a latch

mechanism **44** (described later). The striker entry groove **14** is formed as part of a cover plate **16**. A body **18** is disposed around the cover plate **16**. A vehicle inner side and a vehicle rear side of the latch mechanism **44** are covered by the cover plate **16** and the body **18**.

(22) The door latch device **10** is covered by a case **20**, a first cover **22**, and a second cover **24** in addition to the cover plate **16** and the body **18**. The case **20** mainly covers a vehicle outer side, the first cover **22** mainly covers the vehicle inner side, and the second cover **24** further covers a front upper part on the vehicle inner side of the case **20**. The cover plate **16**, the body **18**, the case **20**, the first cover **22**, and the second cover **24** constitute a housing of the door latch device **10**.

(23) The door latch device **10** further includes a waterproof cover **26** that covers an upper part, a cable cover **28** on a lower part on the vehicle inner side, a coupler **30** disposed on an upper part on the vehicle inner side, and a key cylinder coupling part **32** disposed on an upper part on the vehicle outer side. The waterproof cover **26** covers a boundary part between the case **20** and the first cover **22**, and prevents infiltration of water. The cable cover **28** covers a connecting portion for a cable **35**. The cable **35** links the door latch device **10** with an inner handle (not illustrated). The coupler **30** is connected to a harness connector (not illustrated). The key cylinder coupling part **32** is a portion to be operated when a key is inserted therein. On a portion positioned on the vehicle outer side of the door latch device **10**, an end part of an outer lever **34** connected to an outer handle (not illustrated) is disposed to project toward the outside.

(24) As illustrated in FIG. **3**, an accommodation space **36** is formed on the vehicle inner side of the door latch device **10**. The accommodation space **36** is a region in which the vehicle outer side is covered by the case **20**, and the vehicle inner side is mainly covered by the first cover **22**. The vehicle inner side of the accommodation space **36** is also covered by the cover plate **16**, the body **18**, and the cable cover **28** in addition to the first cover **22**.

(25) The accommodation space **36** briefly includes a mechanism region **40** in which the machine mechanism **38** is disposed, and an electric component region **42** in which electric components are disposed. The electric component region **42** occupies an upper part on the vehicle front side, and the mechanism region **40** occupies a residual portion. The machine mechanism **38** houses the latch mechanism **44** that latches and unlatches the striker by the latch member **12**, and a lock mechanism **46** that is switched between a locked state and an unlocked state. The latch mechanism **44** is disposed on the vehicle rear side in the accommodation space **36**, and covered by the cover plate **16** and the body **18**.

(26) The machine mechanism **38** houses an electric release unit that can release, by power of a motor **94**, a latched state of the striker by the latch mechanism **44**, and a manual release unit that can release, by manual operation, the latched state of the striker by the latch mechanism **44**. The electric release unit includes the motor **94**, a cam wheel (rotating body) **76**, and the like (described later) to unlatch the striker. The manual release unit unlatches the striker via an inner lever **59** (described later) and the outer lever **34** mechanically linked with a manual operation.

(27) As illustrated in FIG. **4**, the latch mechanism **44** includes a base bracket **50**, a ratchet **52**, a ratchet holder **54**, a ratchet lever **56**, an anti-panic lever **58**, and the inner lever **59** in addition to the latch member **12** and the outer lever **34** described above. The respective elements of the latch mechanism **44** are supported by the base bracket **50**.

(28) The latch member **12** is supported by the housing via a latch shaft **60** in a rotatable manner, and includes a striker engagement groove **12a** and a ratchet engagement part **12b**. When the striker enters the striker engagement groove **12a** in accordance with a door-closing operation from a door-opened state, the latch member **12** rotates against elastic force of a spring (not illustrated), and the ratchet **52** engages with the ratchet engagement part **12b** to latch the striker at a full-latch position.

(29) The ratchet **52** includes a base lever **64** supported by the housing via a ratchet shaft **62** in a rotatable manner, and a pole lever **66** supported by the base lever **64** via a base shaft part **66a** in a rotatable manner. The base lever **64** is elastically energized by a base spring **65**. The pole lever **66** is bent in a predetermined angle range with respect to the base lever **64**. A substantially straight

attitude or the ratchet **52** is held by being supported by the ratchet holder **54** from a lateral side, and a distal end of the pole lever **66** engages with the ratchet engagement part **12b** to hold the latch member **12** at the full latch position.

(30) The ratchet holder **54** is supported by the housing via a shaft part **68** in a rotatable manner, and elastically energized by a holder spring **70** to support a lateral side of the base lever **64**. The ratchet holder **54** rotates against elastic force of the holder spring **70** when the ratchet lever **56** rotates, and is separated from the base lever **64**. When the ratchet holder **54** is separated from the base lever **64**, the base lever **64** and the pole lever **66** of the ratchet **52** are caused to be in a buckling state centered on the base shaft part **66a**, and the pole lever **66** leaves the ratchet engagement part **12b** to open the latch member **12**. As a result, the latch member **12** is rotated by elastic force of a spring (not illustrated), and the striker is unlatched. In a case of operating the ratchet **52** via the ratchet holder **54**, the operation can be performed with smaller force as compared with a case of directly operating the ratchet **52**.

(31) The ratchet lever **56** is supported by the base bracket **50** in a rotatable manner, and includes a passive part **56a** projecting toward the vehicle inner side with respect to a rotation axis and a rotation working part **56b** projecting toward the vehicle outer side with respect to the rotation axis. In the ratchet lever **56**, the rotation working part **56b** causes the ratchet holder **54** to rotate when the passive part **56a** moves toward the upper side.

(32) The outer lever **34** is supported by the housing via a shaft part **72** in a rotatable manner, and includes a handle operation part **34a** projecting toward the vehicle outer side with respect to the shaft part **72**, and a working part **34b** and a lever passive piece **34c** projecting toward the vehicle inner side with respect to the shaft part **72**. The handle operation part **34a** is a portion operated by the outer handle. The working part **34b** is inserted into a hole **58a** of the anti-panic lever **58** and an odd-form hole **80b** of an open link **80** (described later). The lever passive piece **34c** is disposed on a lower side of the working part **34b**, and operated by the inner lever **59**. The outer lever **34** is rotated by an operation of the handle operation part **34a** or the lever passive piece **34c**, and pushes up the anti-panic lever **58**.

(33) The inner lever **59** is supported by the housing via the shaft part **74** in a rotatable manner, and is rotated by an operation of the cable **35**. Accordingly, an operation piece **59a** pushes up the lever passive piece **34c**.

(34) The anti-panic lever **58** includes the hole **58a** into which the working part **34b** is inserted, and a working piece **56b** that is bent at an upper part. The anti-panic lever **58** is pushed up by the working part **34b** when the open link **80** (described later) is at an unlocked position and the outer lever **34** rotates, and the working piece **56b** pushes up the passive part **56a** of the ratchet lever **56**. Accordingly, the ratchet holder **54** and the ratchet **52** perform an unlatch operation. The anti-panic lever **58** has a structure separated from the open link **80** due to an anti-panic mechanism.

(35) As illustrated in FIG. 5 and FIG. 6, the lock mechanism **46** includes a cam wheel **76** supported by the housing via a shaft part **76a** in a rotatable manner, a knob lever (first lever) **78** that is supported by the housing via a shaft part **78a** in a rotatable manner and driven by the cam wheel **76**, the open link **80** that is driven by the knob lever **78**, a sub-lock lever **82** linked with the open link **80**, and an open lever (second lever) **84** that is supported by the housing via a shaft part **84a** in a rotatable manner and driven by the cam wheel **76**. The lock mechanism **46** further includes a lock lever **86** and an auxiliary lever **88** linked with the sub-lock lever **82**, and a key lever **90** and a sub-key lever **92** that are linked with a key operation and drive the sub-lock lever **82**. To easily identify components in the respective drawings, the lock lever **86** is represented by fine dots, and the open link **80** is represented by coarse dots.

(36) The cam wheel **76** has a disk shape having teeth (not illustrated) on an outer peripheral surface, meshes with a worm **94a** disposed on a drive shaft of the motor **94** via the teeth on the outer peripheral surface, and can rotate in a case in which the motor **94** is driven. In the following description, for convenience' sake, a case in which the cam wheel **76** rotates clockwise in FIG. 5 is

assumed to be normal rotation, and a case in which the cam wheel **76** rotates counterclockwise is assumed to be reverse rotation.

(37) A cam part **110** is disposed on the cam wheel **76**. The cam part **110** is disposed only on an end face **76b** positioned on the vehicle inner side of the cam wheel **76** (hereinafter, simply referred to as an inner end face **76b**), and projects toward the vehicle inner side. A first cam **111** and a second cam **112** are disposed on the cam part **110**. Cam surfaces **111** and **112a** having different shapes are disposed on respective outer circumferences of the first cam **111** and the second cam **112**. Each of the cam surfaces **111a** and **112a** is formed so that a distance from the center of the cam wheel **76** is increased when the cam wheel **76** rotates counterclockwise.

(38) More specifically, as illustrated in FIG. 5 and FIGS. 7A and 7B, regarding the cam surface **111a** of the first cam **111** positioned on a side close to the inner end face **76b** of the cam wheel **76** in the cam part **110**, a minimum outer diameter portion close to a center part of the cam wheel **76** is assumed to be a starting end **111a1**, and the cam surface **111a** extends in a direction of rotating counterclockwise in FIGS. 7A and 7B. In the example illustrated in FIGS. 7A and 7B, the cam surface **111a** of the first cam **111** is disposed in a range of substantially 270 degrees from the starting end **111a1**. As is clear from FIGS. 7A and 7B, the cam surface **111a** of the first cam **111** is configured to have a distance from the center that is gradually increased in a range of substantially 100 degrees from the starting end **111a1**, and to be a circular arc having a fixed outer diameter after the range of 100 degrees. A maximum outer diameter of the cam surface **111a** of the first cam **111** is set at a distance with which the open lever **84** (described later) can rotate counterclockwise to the maximum in FIG. 5 to perform an unlatch operation on the ratchet holder **54** and the ratchet **52** via the ratchet lever **56**.

(39) On the other hand, regarding the cam surface **112a** of the second cam **112** positioned on a side separated from the inner end face **76b** of the cam wheel **76** in the cam part **110**, a position rotated counterclockwise from the starting end **111a1** of the first cam **111** by substantially 120 degrees is assumed to be a starting end **112a1**, and the cam surface **112a** extends in a direction of rotating counterclockwise in FIGS. 7A and 7B. In the example illustrated in FIGS. 7A and 7B, the cam surface **112a** of the second cam **112** is configured to have a distance from the center that is gradually increased in a range of substantially 100 degrees from the starting end **112a1**, and to be a circular arc having a fixed outer diameter that is the same as that of the cam surface **111a** of the first cam all after the range of 100 degrees. A maximum outer diameter of the cam surface **112a** of the second cam **112** is set at a distance with which the knob lever **78** (described later) can rotate counterclockwise to the maximum in FIG. 5 and the open link **80** is caused to have an erect attitude.

(40) The starting end **112a1** of the cam surface **112a** of the second cam **112** is disposed to be closer to an outer peripheral side than the starting end **111a1** of the cam surface **111a** of the first cam **111** is, and an increasing ratio of the outer diameter of the cam surface **112a** of the second cam **112** is smaller than an increasing ratio of the outer diameter of the cam surface **111a** of the first cam **111**.

(41) As illustrated in FIG. 6, the cam wheel **76** includes an auxiliary component **77** on an end face **76c** positioned on the vehicle outer side (hereinafter, simply referred to as an outer end face **76c**). The auxiliary component **77** is integrally attached to the cam wheel **76** in a state of being prevented from relatively rotating. A torsion coil spring (elastic member) **75** is disposed inside a cylindrical part **77a** formed by the auxiliary component **77**. The torsion coil spring **75** is configured to have a cylindrical coil part by winding a metal fine wire, one spring end part is engaged with the cam wheel **76**, and the other spring end part is engaged with the case **20**. The torsion coil spring **75** allows the cam wheel **76** to normally rotate and reversely rotate with respect to the case **20** in a case in which external force is applied to the cam wheel **76**, while energizing the cam wheel **76** to be maintained at a neutral reference position in a case in which external force is removed.

(42) The auxiliary component **77** includes a regulating projection **77b** projecting toward the vehicle inner side in the vicinity of an outer circumference, and a first tilted wall **77c** that is disposed on a

substantially opposite side of the regulating projection **77b**. The regulating projection **77b** abuts on an elastic member **96a** of a rotation stopper **96** disposed in the case **20** to regulate rotation of the cam wheel **76** when the cam wheel **76** reversely rotates. The first tilted wall **77c** is tilted from an outer peripheral surface of the cylindrical part **77a** toward an outer peripheral side to be gradually positioned on a clockwise direction side in FIG. **6**. The first tilted wall **77c** is disposed to secure a gap between itself and the outer end face **76c** of the cam wheel **76**.

(43) The cam wheel **76** further includes a second tilted wall **76d** and a holding wall **76e**. The second tilted wall **76d** is tilted from the outer peripheral surface of the cylindrical part **77a** of the auxiliary component **77** toward the outer peripheral side to be gradually positioned on a counterclockwise direction side in FIG. **6**. The first tilted wall **77c** and the second tilted wall **76d** extend to be gradually separated from each other from a position close to the cylindrical part **77a** toward the outer peripheral side. The first tilted wall **77c** of the auxiliary component **77** is disposed to be closer to the vehicle outer side than the second tilted wall **76d** is. The holding wall **76e** is a circular arc wall extending from a portion positioned on the outer peripheral side of the second tilted wall **76d** toward the counterclockwise direction in FIG. **6**. As illustrated in FIG. **6**, an end part on the clockwise direction side of the holding wall **76e** is closed by a portion constituting the second tilted wall **76d**, while an end part on the counterclockwise direction side thereof is adjacent to a notch **76f** opening on the outer peripheral surface of the cam wheel **76**.

(44) As illustrated in FIG. **5**, a follower surface **78d** that abuts on the cam surface **112a** of the second cam **112** in the cam part **110** is disposed on the knob lever **78**. The follower surface **78d** is driven by the second cam **112** when the cam wheel **76** normally rotates, and functions to cause the knob lever **78** to rotate counterclockwise against a lever spring **78b**. A knob **78c** is disposed at a distal end part of the knob lever **78**. The knob **78c** engages with a side surface guide groove **80a** disposed on the open link **80**. The knob lever **78** functions such that, when being rotated counterclockwise in FIG. **5**, the knob **78c** moves toward the upper side along the side surface guide groove **80a** to erect the tilted open link **80**, and when being rotated clockwise in FIG. **5**, the knob **78c** moves toward a lower side along the side surface guide groove **80a** to cause the erect open link **80** to be tilted toward the vehicle front side.

(45) The open link **80** includes the odd-form hole **80b** at a lower end, and rotates about the lower end as a center to be switched between an erect attitude (unlocked position) and a tilted attitude (locked position) as described above. The lock mechanism **46** is caused to be in the locked state in a case in which the open link **80** is arranged at the locked position illustrated in FIG. **5**, and the lock mechanism **46** is caused to be in the unlocked state in a case in which the open link **80** is arranged at the unlocked position. That is, when the open link **80** is at the locked position, the anti-panic lever **58** is tilted together with the open link **80**, so that the anti-panic lever **58** does not abut on the ratchet lever **56** even in a case of being lifted up by the outer lever **34**. Thus, the ratchet lever **56** does not operate, and the door is kept closed in the locked state. On the other hand, when the open link **80** is at the unlocked position, the anti-panic lever **58** erects together with the open link **80**, so that the anti-panic lever **58** abuts on and pushes up the ratchet lever **56** when being lifted up by the outer lever **34**. Accordingly, the ratchet lever **56** operates, and the unlocked state in which the door can be opened is caused.

(46) The working part **34b** of the outer lever **34** is inserted into the odd-form hole **80b** of the open link **80**. In a case in which the outer lever **34** operates, the open link **80** moves along an upper and lower direction. The anti-panic lever **58** is assembled to the lower end part of the open link **80**. The anti-panic lever **58** moves integrally with the open link **80**.

(47) The sub-lock lever **82** is supported by the housing via a shaft part **82a** in a rotatable manner, engaged with the lock lever **86** via an outer knob **86c** of the lock lever **86**, and engaged with the open link **80** arranged at the locked position via an inner knob **86i**. That is, when the sub-lock lever **82** rotates counterclockwise in FIG. **5** due to rotation of the key lever **90** and the sub-key lever **92**, the lock lever **86** rotates clockwise in FIG. **5** via the outer knob **86c**, and the lock lever **86** is pushed

out by the inner knob **86i** to cause the open link **80** to be at the unlocked position. In a case in which the sub-lock lever **82** rotates clockwise and returns to an original position, the open link **80** is returned to the locked position by the knob lever **78** that rotates clockwise in FIG. 5 by elastic force of the lever spring **78b**.

(48) The open lever **84** is a constituent element of the electric release unit. That is, the open lever **84** operates to open the door in a case in which the motor **94** is driven by a switch operation and the like performed by a driver, and includes a cam follower part **84b** extending from the shaft part **84a** toward the vehicle front side and a ratchet operation part **84c** extending toward the vehicle rear side. An open spring **84d** is disposed between the open lever **84** and the housing. The open spring **84d** energizes the cam follower part **84b** in the clockwise direction in FIG. 5 to abut on the cam surface **111a** of the first cam **111** in the cam part **110** of the cam wheel **76**. In a case in which the cam wheel **76** normally rotates in FIG. 5, the first cam **111** pushes down the cam follower part **84b**, the open lever **84** rotates counterclockwise about the shaft part **84a** against the open spring **84d**, and the ratchet operation part **84c** moves upward. When the ratchet operation part **84c** moves upward, the passive part **56a** of the ratchet lever **56** is pushed up, the latch mechanism **44** is unlatched, and the door is opened. When the car's wheel **76** returns to the neutral reference position, the open lever **84** also returns to an original attitude by the open spring **84d**.

(49) The open lever **84** can operate the ratchet lever **56** independently of the open link **80**. Thus, with the door latch device **10** including the open lever **84**, the door can be opened by the electric release unit even when the lock mechanism **46** is in the locked state (that is, the open link **80** is at the locked position).

(50) As illustrated in FIG. 6, the lock lever **86** includes an arm **86b** that is supported by the housing in a rotatable manner via a shaft part **86a** and extends toward the upper side, the outer knob **86c** projecting from a distal end of the arm **86b** toward the vehicle outer side, a downward extending part **86d** extending toward the lower side, a first projection **86e** projecting from the downward extending part **86d** toward the vehicle front side, a second projection **86f** projecting from the vicinity of the shaft part **86a** of the arm **86b** toward the vehicle front side, a spring reception part **86g** projecting from the downward extending part **86d** toward the vehicle outer side, and two push-out parts **86h** projecting from a surface on the vehicle inner side of the downward extending part **86d** toward the inner side of the vehicle. The outer knob **86c** is engaged with the sub-lock lever **82** by being fit into a guide hole **82b** formed at the lower end of the sub-lock lever **82**. The lock lever **86** is rotated via the outer knob **86c** when the sub-lock lever **82** rotates, and can be displaced between a working position for switching the open link **80** from the locked position to the unlocked position, and a non-working position not working on the open link **80**.

(51) The spring reception part **86g** abuts on a bent part **100a** of a lock spring **100**. The lock spring **100** defines an attitude of the sub-lock lever **82** via the spring reception part **86g**.

(52) The first projection **86e** abuts on the first tilted wall **77c** in a case in which the cam wheel **76** rotates counterclockwise in FIG. 6. When the cam wheel **76** continuously rotates counterclockwise in a state in which the first tilted wall **77c** abuts on the first projection **86e**, the lock lever **86** rotates clockwise in FIG. 6. The second projection **86f** abuts on the second tilted wall **76d** in a case in which the cam wheel **76** rotates clockwise in FIG. 6. When the cam wheel **76** continuously rotates clockwise in a state in which the second tilted wall **76d** abuts on the second projection **86f**, the lock lever **86** rotates counterclockwise in FIG. 6. The second projection **86f** is disposed to be closer to the vehicle inner side than the first tilted wall **77c** is. The second projection **86f** enters a gap between the first tilted wall **77c** and the outer end face **76c** of the cam wheel **76**, and does not abut on the first tilted wall **77c**. The two push-out parts **86h** abut on the auxiliary lever **88** in a case in which the lock lever **86** rotates clockwise in FIG. 5.

(53) As illustrated in FIG. 5, the auxiliary lever **88** is supported by the shaft part **86a** of the lock lever **86** in a rotatable manner, and includes an arm **88a** projecting from the shaft part **86a** toward the vehicle front side, and a circular arc projection **88b** disposed on an upper part of a distal end of

the arm **88a**. The circular arc projection **88b** abuts on the holding wall **76e** of the cam wheel **76** in a case of rotating counterclockwise in FIG. **6**. An auxiliary lever spring **88c** is interposed between the auxiliary lever **88** and the lock lever **86**, the auxiliary lever spring **88c** energizing the auxiliary lever **88** in a direction of rotating counterclockwise in FIG. **5** with respect to the lock lever **86**. A lower surface of the auxiliary lever **88** is kept abutting on the push-cut parts **86h** due to elastic force of the auxiliary lever spring **88c**.

(54) FIGS. **8A** and **8B** are diagrams illustrating an operation of the lock mechanism **46** viewed from the vehicle inner side when the cam wheel **76** normally rotates (rotates clockwise in FIGS. **8A** and **8B**). The following describes the operation of the lock mechanism **46** with reference to these drawings as appropriate.

(55) As illustrated in FIG. **8A**, in a basic state in which the cam wheel **76** is at the reference position, the cam surface **112a** of the second cam **112** is separated from the follower surface **78d** of the knob lever **78**, and the cam follower part **84b** is separated from the cam surface **111a** of the first cam **111** at a position where the open lever **84** rotates clockwise to the maximum. At this point, the open link **80** is arranged at the locked position, and the knob **78c** of the knob lever **78** is positioned on a lower side of the side surface guide groove **80a**.

(56) When the cam wheel **76** starts to normally rotate by being driven by the motor **94** from the basic state, a portion in the vicinity of the starting end **112a1** of the cam surface **112a** of the second cam **112** abuts on the follower surface **78d** of the knob lever **78**, and the knob lever **78** starts to rotate counterclockwise in FIGS. **8A** and **8B**. Substantially at the same time, a portion in the vicinity of the starting end **111a1** of the cam surface **111a** of the first cam **111** abuts on the cam follower part **84b** of the open lever **84**, and the open lever **84** starts to rotate counterclockwise in FIGS. **8A** and **8B**.

(57) As described above, the increasing ratio of the outer diameter of the cam surface **111a** of the first cam **111** is larger than that of the cam surface **112a** of the second cam **112**. Thus, in a case in which the cam wheel **76** normally rotates, as illustrated in FIG. **8B**, the passive part **56a** of the ratchet lever **56** is sufficiently pushed up by the open lever **84** at an early stage, the latch mechanism **44** unlatches the striker, and the door is opened. On the other hand, the knob lever **78** is caused to be in a state of being rotated counterclockwise to the maximum by the cam surface of the second cam **112**, the open link **80** gradually rotates clockwise when the knob **78c** moves toward the upper side along the side surface guide groove **80a**, and the open link **80** is caused to be at the unlocked position. The sub-lock lever **82**, the lock lever **86**, and the auxiliary lever **88** do not operate in this period, and are maintained in the basic state in FIG. **8A**.

(58) During the operation described above, the torsion coil spring **75** interposed between the case **20** and the cam wheel **76** is gradually bent. Thus, when electric supply to the motor **94** is stopped thereafter, the cam wheel **76** rotates counterclockwise to be at the reference position due to elastic force of the torsion coil spring **75**, and the lock mechanism **46** returns to the basic state illustrated in FIG. **8A**.

(59) At the time of such electric release, as illustrated in FIG. **8A** and FIG. **8B**, the open lever **84** can be rotated by being driven by the motor **94** to work on the latch mechanism **44** to unlatch the striker. At this point, the open link **80** also reciprocates between the locked position and the unlocked position. The open link **80** does not work on the other components, but synchronously operates at appropriate time intervals at the time of electric release, so that it is possible to prevent grease from being hardened due to long-term deterioration, or prevent a spring, a lever, and the like made of steel material from getting rusty. Accordingly, a smooth operation of the lock mechanism **46** can be secured at all times.

(60) At the time of electric release, only the open link **80** synchronously operates, and the lock lever **86** does not operate. Thus, the spring reception part **86g** of the lock lever **86** does not get over the bent part **100a**, and sound is not generated, so that a user does not feel a sense of incongruity.

(61) FIGS. **9A** to **9D** are diagrams illustrating an operation of the lock mechanism **46** viewed from

the vehicle outer side when the cam wheel **76** reversely rotates (rotates clockwise in FIGS. **9A** to **9D**) and normally rotates (rotates counterclockwise in FIGS. **9A** to **9D**). The following further describes the operation of the lock mechanism **46** with reference to the drawings as appropriate.

(62) As illustrated in FIG. **9A**, in the basic state in which the cam wheel **76** is arranged at the reference position, the lock lever **86** is at a position where the lock lever **86** rotates clockwise to the maximum, and the first projection **86e** and the second projection **86f** are in a non-abutting state with respect to the cam wheel **76**.

(63) When the cam wheel **76** starts to reversely rotate by being driven by the motor **94** from the reference state, and rotation of the cam wheel **76** proceeds to substantially 40 degrees as illustrated in FIG. **9B**, the second tilted wall **76d** of the cam wheel **76** abuts on the second projection **86f**. Accordingly, the lock lever **86** rotates counterclockwise, and the spring reception part **86g** is displaced to get over the bent part **100a** of the lock spring **100**.

(64) When the lock lever **86** rotates, the sub-lock lever **82** is rotated clockwise via the outer knob **86c**, the open link **80** is rotated counterclockwise via the inner knob **86i**, and the auxiliary lever **88** is rotated counterclockwise by the push-out parts **86h**. Accordingly, each of the sub-lock lever **82** and the open link **80** is caused to be at the unlocked position, and the circular arc projection **88b** of the auxiliary lever **86** is displaced to a position close to the cylindrical part **77a** via the notch **76f**.

(65) When the spring reception part **86g** gets over the bent part **100a** or the lock spring **100**, the cam wheel **76** starts to normally rotate by being driven by the motor **94**. As illustrated in FIG. **9C**, when normal rotation of the cam wheel **76** proceeds to substantially 40 degrees from the state of FIG. **9B**, the cam wheel **76** returns to the reference position illustrated in FIG. **9A**. However, the spring reception part **86g** is held by the bent part **100a**, so that the lock lever **86**, the sub-lock lever **82**, and the open link **80** are kept in the attitude illustrated in FIG. **9B**. Accordingly, the lock mechanism **46** is caused to be in the unlocked state. At this point, due to normal rotation of the cam wheel **76**, the holding wall **76e** is arranged on an outer peripheral side of the circular arc projection **88b**, and the circular arc projection **88b** is caused to be in a state of engaging with an inner peripheral surface of the holding wall **76e**.

(66) When normal rotation of the cam wheel **76** further proceeds by substantially 40 degrees from the state illustrated in FIG. **9C**, the first tilted wall **77c** of the cam wheel **76** abuts on the first projection **86e**. Accordingly, the lock lever **86** rotates clockwise, and as illustrated in FIG. **9D**, the spring reception part **86g** gets over the bent part **100a** of the lock spring **100** to return to the reference state illustrated in FIG. **9A**.

(67) When the lock lever **86** rotates clockwise, the sub-lock lever **82** is rotated counterclockwise via the outer knob **86c**, the open link **80** is rotated clockwise by the knob lever **78** that is rotated by elastic force of the lever spring **78b**, and each of the sub-lock lever **82** and the open link **80** returns to the reference state illustrated in FIG. **9A**.

(68) On the other hand, the circular arc projection **88b** engages with the holding wall **76e** of the cam wheel **76**, so that the auxiliary lever **88** is maintained in a state of rotating counterclockwise against elastic force of the auxiliary lever spring **88c**. When normal rotation of the cam wheel **76** further proceeds from this state, the regulating projection **77b** abuts on the rotation stopper **96** of the case **20** via the elastic member **96a**, and rotation of the cam wheel **76** is stopped. Due to this, excessive rotation of the cam wheel **76** can be prevented.

(69) After the regulating projection **77b** abuts on the rotation stopper **96** and rotation of the cam wheel **76** is stopped, the cam wheel **76** starts to reversely rotate by being driven by the motor **94**. When reverse rotation of the cam wheel **76** proceeds to the position illustrated in FIG. **9A**, engagement between circular arc projection. **88b** and the holding wall **76e** is released. Due to this, the auxiliary lever **88** is rotated clockwise by elastic force of the auxiliary lever spring **88c**, and returns to the position illustrated in FIG. **9A**. In this way, the entire lock mechanism **46** returns to the basic state illustrated in FIG. **9A**.

(70) In this way, with the door latch device **10** described above, the unlatch operation for the latch

mechanism **44** and switching of the lock mechanism **46** between the locked state and the unlocked state can be performed by the single motor **94**. Furthermore the cam part **110** is disposed only on the one inner end face **76b** of the cam wheel **76**, so that the knob lever **78** and the open lever may be disposed only in a region opposed to the inner end face **76b** of the cam wheel **76**, and the levers **78** and **84** are not disposed in a region opposed to the outer peripheral surface of the cam wheel **76**. Thus, it possible to prevent a situation in which the size of the door latch device **10** is increased.

(71) The individual cam surfaces **111a** and **112a** are caused to abut on the knob lever **78** and the open lever **84**, so that the levers **78** and **84** can be operated at a desired optional timing without influencing mutual operations by appropriately changing profiles of the cam surfaces **111a** and **112a**, and the door latch device **10** satisfying needs of the user can be applied.

(72) In the embodiment described above, the first cam and the second cam are disposed on the outer peripheral surface of the cam part, but the embodiment is not limited thereto. Any one of the cam surfaces or both of the cam surfaces may be formed on the inner peripheral surface of the cam part.

(73) In the embodiment described above, exemplified is the actuator applied to the door latch device, but the actuator can also be applied as an actuator for another device.

(74) According to the present disclosure, the cam part is disposed only on the one end face of the rotating body, so that each of the first lever and the second lever may be disposed only in a region opposed to the one end face of the rotating body, and each lever is not disposed in a region opposed to the outer peripheral surface of the rotating body. Accordingly, it is possible to prevent the size of the device from being increased.

(75) Although the disclosure has been described with respect to specific embodiments for a complete and clear disclosure, the appended claims are not to be thus limited but are to be construed as embodying all modifications and alternative constructions that may occur to one skilled in the art that fairly fall within the basic teaching herein set forth.

Claims

1. A door latch device comprising: a latch mechanism which maintains a door closed relative to a vehicle body by engaging with a striker to bring the striker into a latched state; a motor; a rotating body configured to be rotated and driven by the motor with respect to a case; and a lock mechanism including i) a first lever and a second lever which are disposed to be rotatable with respect to the case and are configured to be operated by a cam part disposed on the rotating body and ii) an open link which moves between a locked position and an unlocked position, wherein the lock mechanism switches between i) a locked state in which the open link is at the locked position such that a releasing operation of the latched state by a manual operation is inactivated and ii) an unlocked state in which the open link is at the unlocked position such that the releasing operation of the latched state by the manual operation is activated, wherein the open link is moved to the unlocked position when the first lever is rotated by the cam part, wherein the latched state of the striker is released when the second lever is rotated by the cam part when the lock mechanism is in the locked state, wherein the cam part is disposed only on one end face of the rotating body, wherein the cam part includes a first cam configured to operate the first lever and a second cam configured to operate the second lever which are individually disposed in the cam part, wherein when the state of the latch mechanism is set to an unlatched state, not only the second lever but also the first lever are rotated so as to rotate the open link, thereby increasing a frequency of stirring grease inside the case by the open link, wherein the first lever includes a knob disposed at a distal end part thereof, wherein the open link includes a side surface guide groove, and wherein the knob of the first lever engages with the side surface guide groove of the open link so that the open link rotates when the knob moves along the side surface guide groove.
2. The door latch device according to claim 1, wherein each of the first cam and the second cam

has a cam surface on an outer circumference, and the cam surface of the first cam and the cam surface of the second cam are disposed side by side with each other in an axial direction of the rotating body.

3. The door latch device according to claim 2, wherein the cam surface of the first cam and the cam surface of the second cam are configured to partially match with each other.

4. The door latch device according to claim 1, wherein an elastic member, which is configured to maintain the rotating body at a neutral position with respect to the case, is disposed between the case and another end face of the rotating body.
